

Homework Assingment 2

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Acknowledgement I accept the code of conduct rules as stated by the professor.

Parameters

Before explaining the code needed to apply bootstrap algorithm, we first choose the necessary parameters.

```
set.seed(123)
theta <- 1
ns <- c(10,20,50,100,200,300,400,500,1000,2000)
B <- 1000
alpha <- 0.05
```

Bootstrap

In order to apply the algorithm, we first need to set two functions which will be computed at each bootstrap replications $b = 1 \dots 1000$. Then we set an empty vector that should contain our final confidence intervals. We will make a loop that will go over all the possible values of sample sizes, and at every one of them we will have 1000 re-samples of the original quasi random sample from $U(0,1)$. At each new re-sample we will compute the values of estimators. This will produce the distribution of the estimators, allowing as to compute 95% confidence interval for each.

```
# Estimators
T_1 <-function(x,i){
  max(x[i])
}

T_2 <-function(x,i){
  ((i+1)/i)*max(x[i])
}

# Storage
results_package_bootstrap <- as.numeric()
for (j in 1:length(ns)) {
  n <- ns[j]
  # Simulating one dataset from Uniform(0, theta)
  x <- runif(n,0, theta)
  # Bootstrap for T1
  boot_T1 <- boot(data = x, statistic = T_1, R = B)
  ci_T1 <- boot.ci(boot_T1,
    conf= 1-alpha,
```

```

                                type = "perc")
# Bootstrap for T2
boot_T2 <- boot(data = x, statistic = T_2, R = B)
ci_T2 <- boot.ci(boot_T2,
                 conf= 1-alpha,
                 type = "perc")
# Store results in a temporary object
results_temp <- c(sample_size = ns[j],
                  t1_lower_bound = round(ci_T1$percent[1,4],3),
                  t1_upper_bound = round(ci_T1$percent[1,5],3),
                  t2_lower_bound = round(ci_T2$percent[1,4],3),
                  t2_upper_bound = round(ci_T2$percent[1,5],3))
results_package_bootstrap <- rbind(results_package_bootstrap,results_temp)
}
rownames(results_package_bootstrap) <- rep('',10)
print(results_package_bootstrap)

```

##	sample_size	t1_lower_bound	t1_upper_bound	t2_lower_bound	t2_upper_bound
##	10	0.883	0.940	0.981	1.881
##	20	0.771	0.995	0.868	1.990
##	50	0.916	0.986	0.947	1.479
##	100	0.953	0.984	0.966	1.312
##	200	0.981	0.991	0.990	1.188
##	300	0.976	0.990	0.987	1.132
##	400	0.986	1.000	0.990	1.091
##	500	0.996	0.998	0.999	1.050
##	1000	0.996	0.998	0.998	1.042
##	2000	0.999	0.999	0.999	1.020

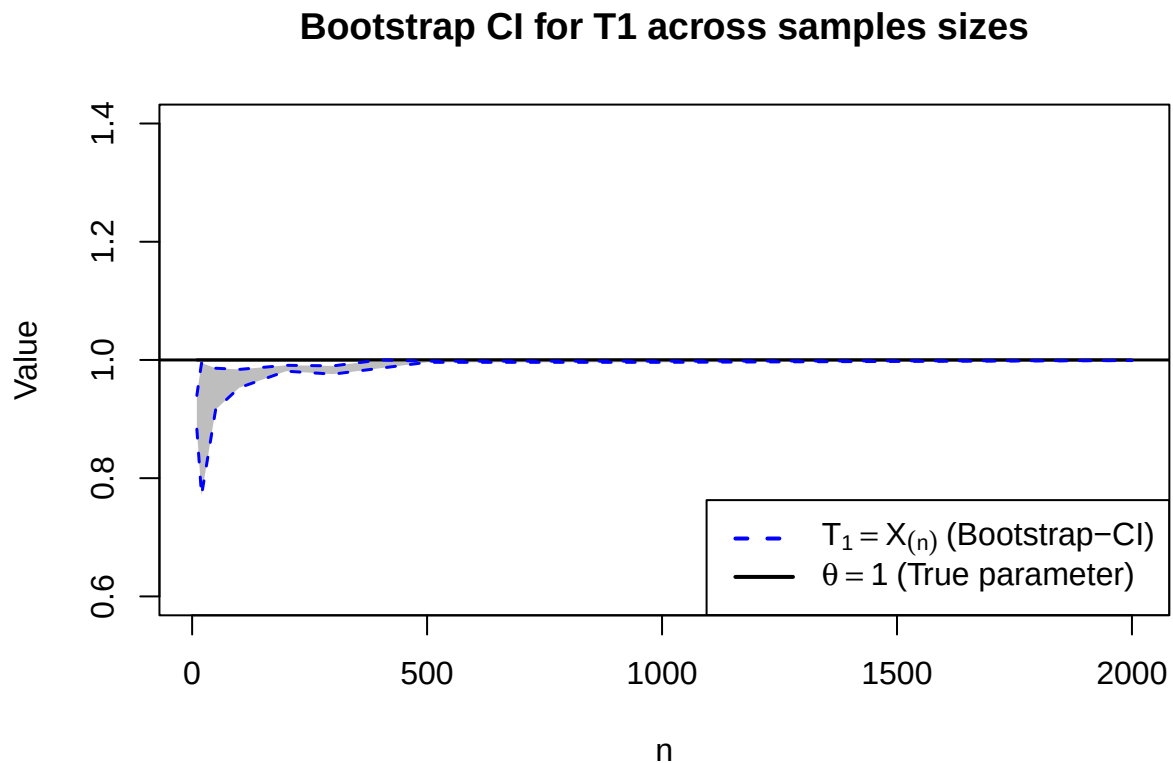
Plots

Finally, we are able to plot the confidence intervals for both estimators:

```
plot(ns, rep(theta,length(ns)), type="l", lwd = 1.5, ylab="Value", xlab="n", main = "Bootstrap CI for T1")
polygon(c(rev(ns),ns),c(rev(results_package_bootstrap[,2]),results_package_bootstrap[,3]), col = "gray",
abline(h = theta, col = "black", lwd = 1.5)

lines(ns, results_package_bootstrap[,2], lty="dashed", lwd = 1.5, col = "blue")
lines(ns, results_package_bootstrap[,3], lty = 'dashed', lwd = 1.5, col = "blue")

legend(x= "bottomright",
      legend = c(
        expression(T[1] == X[(n)] ~ "(Bootstrap-CI)"),
        expression(theta == 1 ~ "(True parameter)"),
      ),
      lty = c(2,1),
      col = c("blue","black"),
      lwd = 2
    )
```



```

plot(ns, rep(theta,length(ns)), type="l", lwd = 1.5, ylab="Value", xlab="n",main = "Bootstrap CI for T2

polygon(c(rev(ns),ns),c(rev(results_package_bootstrap[,4]),results_package_bootstrap[,5]), col ="gray",

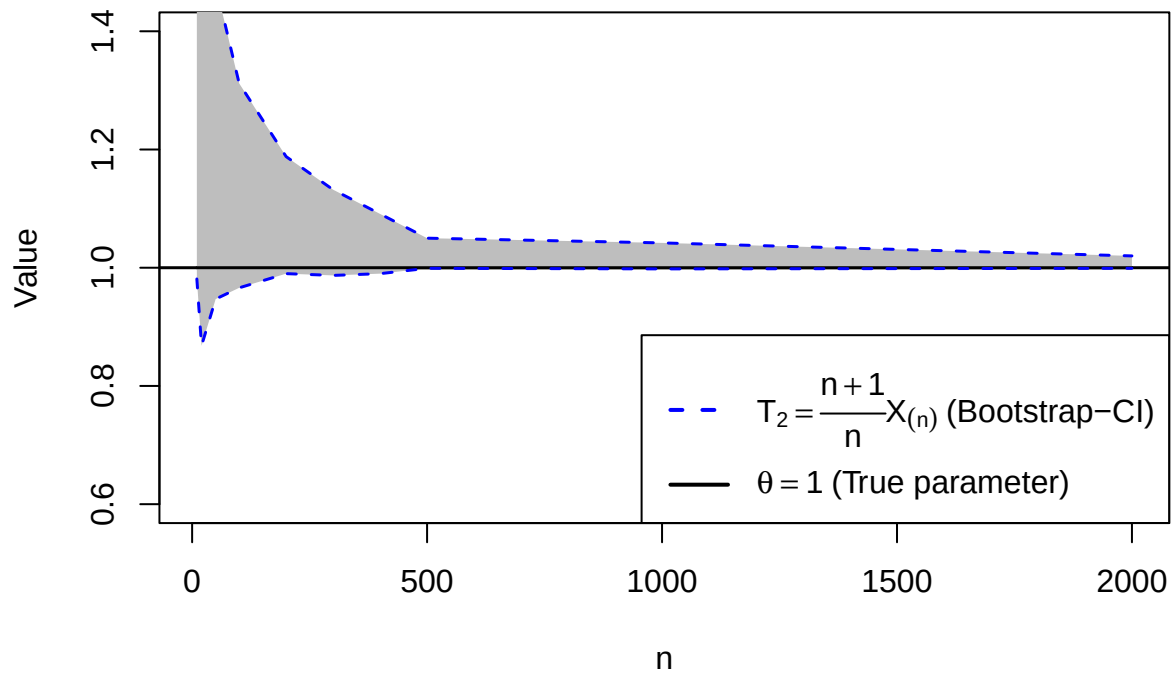
abline(h = theta, col = "black", lwd = 1.5)

lines(ns, results_package_bootstrap[,4], lty = 'dashed', lwd = 1.5, col = "blue")
lines(ns, results_package_bootstrap[,5], lty = 'dashed', lwd = 1.5, col = "blue")

legend(x= "bottomright",
      legend = c(
        expression(T[2] == frac(n+1,n)*X[(n)] ~ "(Bootstrap-CI)"),
        expression(theta == 1 ~ "(True parameter)"),
      ),
      lty = c(2,1),
      col = c("blue","black"),
      lwd = 2
    )

```

Bootstrap CI for T2 across samples sizes



Additional Questions

1. In order to answer the first question, we need to look into asymptotic properties of an estimator. In order for an estimator to be consistent, we need it to be asymptotically unbiased meaning that $\lim_{n \rightarrow \infty} E[\hat{\theta}_n - \theta] = 0$. Furthermore if asymptotically unbiased estimator also has $\lim_{n \rightarrow \infty} \text{var}(\hat{\theta}_n) = 0$, then we can say that $\hat{\theta}_n$ is consistent for θ . This could also be seen by taking $\lim_{n \rightarrow \infty} \text{MSE}[\hat{\theta}_n]$. First we set $T_1 = \hat{\theta}_n$, then because we know the density of Uniform random variable, and we know the CDF of maximum order statistics we can conclude that the $\text{Bias}(T_1) = \frac{-1}{n+1}\theta$, which as $n \rightarrow \infty$ is equal to 0. Furthermore $\text{var}(T_1) = \frac{\theta^2 n}{(n+2)(n+1)^2}$ which also goes to 0. Hence, we can conclude that asymptotically MSE goes to 0 and our T_1 is consistent estimator for θ . If we consider the equality such that $T_2 = \frac{n+1}{n}T_1$, we could see that the $\text{MSE}[T_2] = \frac{\theta^2}{n(n+2)}$, which also goes to 0 as $n \rightarrow \infty$. Suggesting that as T_1 , T_2 is also consistent estimator for θ .
2. To run bootstrap we needed to make choices concerning 4 parameters: θ , n , B , α and simulation seed.
 - θ is the population parameter, that is needed for uniform distribution. In this example it is assumed to be known and we set it to 1.
 - n is the sample size at each of our B re-samples, we use different values of n , so we could observe how our parameter differ as n fluctuates.
 - B is the number of samples we draw with replacement from the original sample. We need to set it reasonably high, so the asymptotic properties are visible, as well for our results to be reliable and stable.
 - α is needed so we could set the confidence bands, we set it to 5%, so the coverage probability of the true value of θ is 95%.
 - Simulation seed, is set by default 123, it is needed so we could replicate our findings on which further theoretical questions are based. It is supposed to protect us from unexpected outcomes due to the repeated simulations.
3. As we can see in both graphs the confidence interval converges to the true value, as the value of n diverges. This is expected as, both equalities for lower and upper bound have n in denominator, so the bounds get increasingly tighter around true value as n increases. This is further derived from the fact that both estimators are asymptotically unbiased. Furthermore the values of T_2 are above maximum of the parameter, but that is simply because its support is wider than the one for the uniform distribution, since we are multiplying it by a number larger than 1, (whereas for T_1 the support is the same as for the distribution).
4. For each of our B re-samples, we compute the T_1 statistic (maximum of the sample), and then produce the empirical distribution function of our estimator. Which has the support of the observed values, they can never be higher than the maximum values, so there is no right tail, and the EDF is bias downward. T_2 corrects this bias with the factor $\frac{n+1}{n}$, hence reaching the true value, furthermore this could also be explained by T_2 being unbiased for θ .
5. This is possible, it may happen that confidence interval does not contain the true value, namely the width of the confidence interval directly depends on the probability that the interval contain the certain population parameter, and this probability is never 100% for any finite sample size.
6. From our example we see that as n increases the confidence bands become tighter around the true population parameter. There is too much variability for small sample sizes causing the intervals to be broader. Furthermore the maximum may be too small or too large which will push T_2 in some direction causing the entire bootstrap distribution to miss on the true value of θ . However this could be further affected by the value of θ , because the distribution of our parameters in bootstrap, is based on the sample we draw, and if this sample comes from skewed distribution, it may cause our parameters to be biased, missing the true value completely.
7. One could study this possibility by running Monte Carlo simulation, repeatedly generating datasets from $U(0, \theta)$, compute the bootstrap CI's and checking if and how many do not contain the value of θ .